

Case Study A

Design of Working/Living Environment

Disclaimer: The use case presented below is inspired by a real-world study, in which research teams faced specific data-related challenges from a storage/processing perspective, privacy and compliance with GDPR point of view, or ethical considerations and institutional ethical review

Acknowledgement: this use case was written jointly by data stewards from TU Eindhoven, University of Twente, and TU Delft.

Project Overview

Due to the 2019/2022 pandemic, a growing number of people transitioned to working from home. This project aims to identify the key elements of a combined working and living environment, proposing customised designs tailored to different user groups based on their activities and living/working conditions.

The project involves gathering data on the new work-life physical setups adopted by the general population since the onset of the COVID pandemic. This will be followed by an analysis of how these new arrangements benefit or hinder individuals' lives and work. Finally, the project will propose various spatial organisation solutions, reflecting the diverse work-life setups observed.

A participatory approach will be employed for data collection and the design of these environments. This will involve frequent interactions between researchers and participants. A pivotal aspect of the research is the use of a Virtual Reality (VR) environment, enabling participants to experience and provide feedback on the proposed work-life space designs.

Data Collection

The exploration of current work-life setups will include analysing social media content and collecting images and descriptions of new workspace arrangements, supplemented by a review of existing academic literature. These insights will inform the design of test environments, which participants will evaluate.

A focus group methodology will be employed to gather feedback and reflections on the various interior design proposals. The proposed work-life spaces will be modelled in 3D and hosted on a platform compatible with VR tools, allowing participants to interact with and assess these environments.

The collected data will include:

- Social media images from diverse sources (e.g., Instagram, Pinterest), adhering to applicable copyrights.
- Online focus group session recordings.
- Voice recordings
- Transcriptions of participant interactions for analysis.
- Virtual models of work environments and participant assessments.
- Test results derived from interviews conducted after participants complete the VR experiment.

Data Publishing

The research team plans to share their findings and methodologies through the following channels:

- A public report featuring visuals, such as images of the VR setup, to illustrate the evaluation process of the virtual work environments.
- Public displays showcasing the work-life environment designs considered for creating optimal spaces.
- Public presentations of the finalised work-life environment designs developed and tested during the study.